10TH CLASS MATHEMATICS — NEW SYLLABUS (PECTAA)

Exercise 7.1 — Signs and Values of Trigonometric Ratios

Unit 7 | Solved Exercise | New Book (PECTAA)

Exercise 7.1 (Solved)

Every angle is first written as 180° − θ, 90° + θ or 180° + θ, and then the quadrant decides the sign. In the first quadrant all ratios are positive, in the second only sin and cosec are positive.

Signs of the trigonometric functions

Quadrant	Angle	Positive ratios
I	0° — 90°	all
II	90° — 180°	sin, cosec
III	180° — 270°	tan, cot
IV	270° — 360°	cos, sec

1. Find the signs of the following:

(i) sin 55° (ii) cos 145° (iii) tan 111°

(iv) sec 179° (v) cosec 88° (vi) cot 14°

(i) sin 55°

55° lies in the first quadrant

Answer: sin 55° is positive

(ii) cos 145°

145° lies in the second quadrant, cos is negative there

Answer: cos 145° is negative

(iii) tan 111°

111° lies in the second quadrant, tan is negative there

Answer: tan 111° is negative

(iv) sec 179°

179° lies in the second quadrant, sec is negative there

Answer: sec 179° is negative

(v) cosec 88°

88° lies in the first quadrant

Answer: cosec 88° is positive

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(vi) cot 14°

14° lies in the first quadrant

Answer: cot 14° is positive

2. Fill in the blanks:

Allied angles

Angle	Value	Reason
(i) tan(180° − θ)	− tan θ	second quadrant, tan negative
(ii) sin(180° − θ)	+ sin θ	second quadrant, sin positive
(iii) tan(90° + θ)	− cot θ	second quadrant, tan negative
(iv) cos(90° + θ)	− sin θ	second quadrant, cos negative
(v) cosec(180° − θ)	+ cosec θ	second quadrant, cosec positive
(vi) sec(90° − θ)	+ cosec θ	sec of the complement is cosec

3. Without using calculator, find the exact values of the following trigonometric functions.

(i) sin 150° (ii) tan 150° (iii) sec 150°

(iv) cosec 120° (v) cos 120° (vi) cot 120°

(vii) sin 135° (viii) sec 135° (ix) cot 135°

(i) sin 150°

= sin(180° − 30°) = sin 30°

So sin 150° = $\frac{1}{2}$

Answer: sin 150° = 1/2

(ii) tan 150°

= tan(180° − 30°) = − tan 30°

So tan 150° = − $\frac{1}{\sqrt{3}}$

Answer: tan 150° = −1/√3

(iii) sec 150°

= sec(180° − 30°) = − sec 30°

So sec 150° = − $\frac{2}{\sqrt{3}}$

Answer: sec 150° = −2/√3

(iv) cosec 120°

= cosec(180° − 60°) = cosec 60°

So cosec 120° = $\frac{2}{\sqrt{3}}$

Answer: cosec 120° = 2/√3

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(v) cos 120°

$= \cos(180° - 60°) = -\cos 60°$

So $\cos 120° = -\frac{1}{2}$

Answer: cos 120° = −1/2

(vi) cot 120°

$= \cot(180° - 60°) = -\cot 60°$

So $\cot 120° = -\frac{1}{\sqrt{3}}$

Answer: cot 120° = −1/√3

(vii) sin 135°

$= \sin(180° - 45°) = \sin 45°$

So $\sin 135° = \frac{1}{\sqrt{2}}$

Answer: sin 135° = 1/√2

(viii) sec 135°

$= \sec(180° - 45°) = -\sec 45°$

So $\sec 135° = -\sqrt{2}$

Answer: sec 135° = −√2

(ix) cot 135°

$= \cot(180° - 45°) = -\cot 45°$

So $\cot 135° = -1$

Answer: cot 135° = −1

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